

# Pure Arterial Malformation (PAM): Case Report and Review of Literature

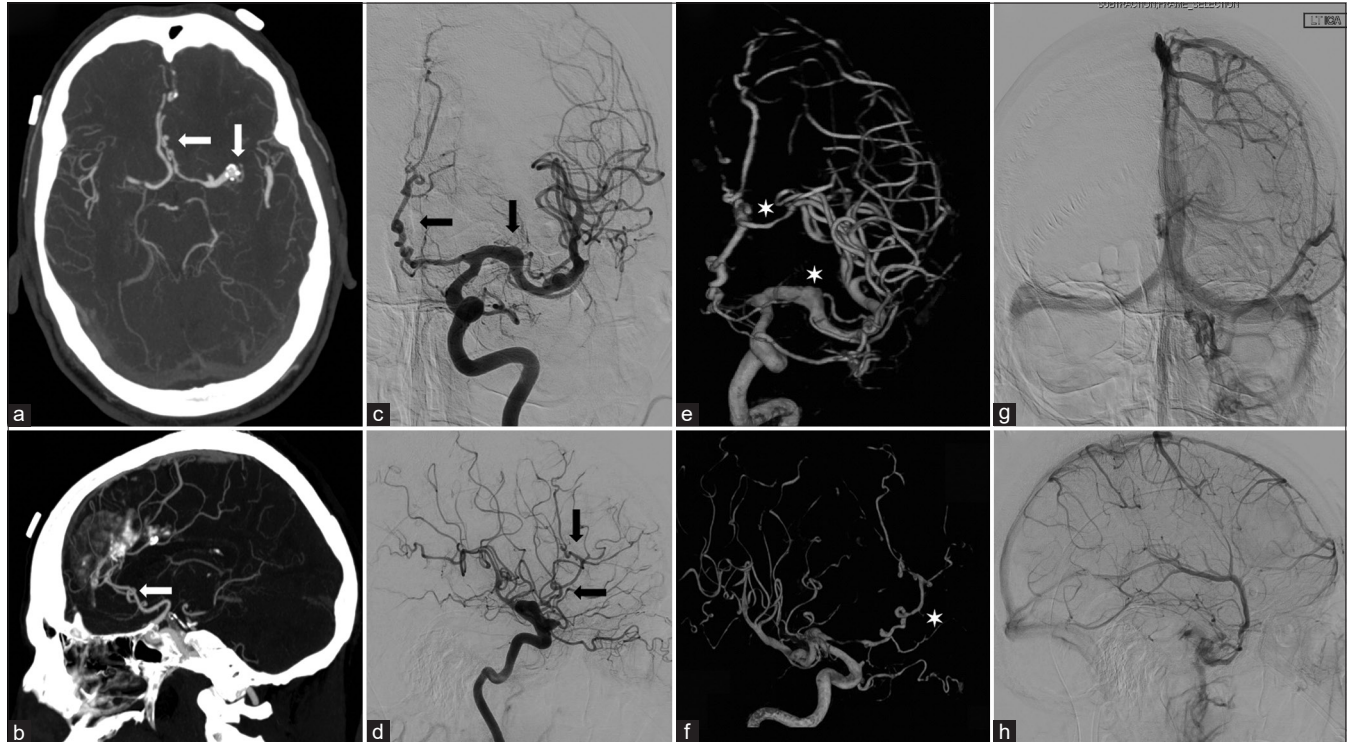
Dear Editor,

Pure arterial malformations (PAMs) are defined as dilated, overlapping, and tortuous arteries and/or a mass of arterial loops without arteriovenous (AV) shunting. The term pure arterial malformation (PAM) was first described in 2013 by McLaughlin *et al.*,<sup>[1]</sup> and since then only a few cases have been reported. The etiology, natural history, and treatment strategies of PAM have not been fully elucidated. We describe a case of PAM involving the left anterior cerebral anterior cerebral circulation (ACA and MCA).

A male in his mid-sixties underwent vascular imaging for a recently diagnosed left medial frontal and left occipital meningiomas. Cranial CT angiography (CTA) and digital subtraction angiography (DSA) demonstrated vascular abnormalities involving the left ACA and MCA. The left ACA showed a dysplastic, dilated, and tortuous vessel with a redundant A2 and proximal A3 segment of the left ACA, angioarchitecture consistent with a PAM [Figure 1]. The left ACA showed a mass of arterial loops without evidence of AV shunting or aneurysmal dilatation. In addition, the proximal

M1 segment and the M2 bifurcation segments are dysplastic with surrounding calcifications [Figure 1].

Pure arterial malformations are a recently described intracranial vascular abnormality that are commonly mistaken for aneurysms or arteriovenous malformations (AVM) on non-invasive imaging. They are characterized by the absence of any venous component, thus, clearly differentiating PAM intracranial vascular malformations with AV shunting like AVMs. In addition, PAM can be erroneously labeled as a dolichoectasia. But dolichoectasia predominantly involves the vertebrobasilar system and/or internal cerebral artery and rarely involves distal vessels.<sup>[2]</sup> Dolichoectatic vessels are elongated, dilated, and sometimes tortuous, although they are still recognizable, whereas with PAM the tortuosity is often so severe that the vessel has the appearance of a mass or cluster of arterial loops. The pathomechanism of dolichoectasia can be divided into the following subtypes: atherosclerotic, non-atherosclerotic, or associated with an arterial dissection. The vast majority of dolichoectasia are due to atherosclerotic, and these patients tend to be older males with a history of vascular comorbidities such as hypertension and smoking.



**Figure 1:** Representative case with cranial CTA and DSA demonstrating a pure arterial malformation involving the left anterior cerebral circulation. (a and b) CTA axial and sagittal (MIP) image showing tortuosity and arterial loops involving the left A2 ACA, proximal A3, and left M1 segment of the MCA with surrounding calcification (white arrow). (c-f) 2D DSA image (AP and lateral view, black arrow), and 3D DSA image (AP and Lateral view, Star sign \*) showing pure arterial malformation involving ACA and MCA. (g and h) Venous phase of 2D DSA (AP and lateral view) showing normal venous draining pattern

The non-atherosclerotic variants are typically seen in younger patients with underlying connective tissue disorders.<sup>[2]</sup> The incidence of PAM is very low with few cases reported in the literature<sup>[3,4]</sup> and typically seen in young adult females, with male-to-female ratio of 1:3.<sup>[4]</sup> The majority of PAM cases are clinically asymptomatic and reported as an incidental finding with only a single case report of a symptomatic PAM.<sup>[4]</sup> Although any intracranial vessels can be involved, there is a predilection for anterior cerebral artery (ACA) and posterior cerebral artery (PCA).<sup>[5]</sup> Involvement of the ACA involves distal A2 and A3 segments with surrounding calcification.<sup>[5]</sup> Catheter angiography is essential for the diagnosis as it provides the temporal and spatial resolution necessary to exclude AV shunting and to discern the angioarchitecture of the tightly coiled arterial loops.

The exact etiology of pure arterial malformations is unknown; however, possible pathogenic mechanisms include congenital insult or somatic mutation later in life, underlying connective tissue disorder, post-infectious and/or a chronic healed arterial dissection.<sup>[5]</sup> Cortical or white matter dysplasia in the area supplied by the PAM has been noted in the literature pointing toward a congenital etiology, at least in a few cases detected in childhood.<sup>[6]</sup> Chronic calcifications associated with PAM suggest a long-standing pathological process prior to the ultimate diagnosis. Indeed, early childhood viral injection has been implicated as a possible etiology of PAM.<sup>[5,7]</sup>

Our case showed PAM involving the left anterior cerebral circulation. The associated calcifications would suggest a long-standing lesion that likely occurred at a much younger age. Similar to the majority of cases published to date, the PAM in our patient was clinically asymptomatic and he underwent successful surgical treatment of his meningioma. Unlike other malformations, PAM has a more benign natural history and is generally managed conservatively. We believe that pure arterial malformations are likely more common than reported, and their understanding is important for diagnosis and appropriate patient counseling. Further, studies on the etiologies, predisposing factors, genetic association, and treatment strategies for PAMs are needed.

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## Conflicts of interest

There are no conflicts of interest.

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